



CS353 Project Proposal

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Group 18

Berra Alkış - 22302527 – Section 2

Doğa Ertürk - 22202567 – Section 2

Yasemin Altun – 22202739 – Section 1

Melike Feyza Bahadırlı - 22201729 – Section 1

Ayşe Tayyibe Zehra Özkan - 22301879 – Section 1

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1.1 Problem Description

Overview and Purpose

This project, Veterinary Clinic Chain Management System, is a web-based database application that supports a multi-branch veterinary clinic network. The system provides a structured, secure and reliable environment where different user roles interact through pet profiles, appointments, medical visit records, prescriptions, inventory operations, billing, and vaccination tracking. The main goal of this platform is to unify clinical and operational workflows, so that pet owners can access services in a more convenient and easier way, veterinarians can document care consistently, and managers/administrators can monitor branch operations with centralized oversight.

Unlike a single-clinic solution, this system is designed to manage a chain of clinics where each branch has its own schedule and stock, while the organization needs consistent patient records and operational rules across branches. This requires both management of branch-based data such as appointments and inventory per branch as well as chain-level consistency, continuity of a pet's medical timeline, and referral to other specialists.

The system is built around three primary actors: pet owners, veterinarians, and managers/administrators.

- **Pet owners** can register and manage their pets (species, breed, age, allergies), browse available branches and veterinarians using filters such as branch location, specialization, species expertise, and available dates, and book appointments by selecting a veterinarian and a time slot. During booking, the system enforces key rules by checking scheduling conflicts, the veterinarian's maximum daily appointment limit, and whether the pet has any outstanding unpaid bills. After completion, owners can view visit summaries and rate veterinarians.
- **Veterinarians** can open scheduled appointments and view the pet's full medical history, including past diagnoses, prescriptions, vaccinations, and allergies. They document the current visit by recording diagnosis, symptoms, treatment decisions, prescriptions and follow-up notes. If medication is prescribed, the system checks the clinic inventory for availability and deducts stock accordingly. When specialized care is required at another branch, veterinarians can create a referral record linked to the pet and the target branch/specialist to consult the specialist while ensuring continuity of care.
- **Managers/administrators** supervise operational processes such as tracking and monitoring stocks and shipments across different branches, primarily focusing on inventory visibility and compliance-oriented monitoring. They can follow system outputs such as stock usage, driven by prescriptions; billing-related constraints, and vaccination compliance indicators derived from stored due dates and records. This supports consistent management while preserving branch-level operations.

In the primary workflow, an owner selects a branch and a veterinarian, chooses an available time slot, and the system validates the booking based on availability and blocking conditions including availability of the veterinarian and the veterinarian's maximum daily appointment limit. During the appointment, the veterinarian records clinical details and may prescribe medication, which triggers inventory checks and deductions. If needed, a referral is created for another branch. After the appointment is completed, the system generates a billing record. Finally, the owner can see the bill generated for them, review the visit summary and submit a veterinarian rating, while the pet's medical timeline is updated with database support to enforce slot limits, stock deductions, and billing generation.

The system also introduces an extended feature: **Pet Microchip & Lost/Found Registry**. This module allows registering a unique microchip for a pet and creating lost/found reports across branches to help reunite pets with owners. This extension enriches the database design by introducing new entities and relationships and cross-branch lost/found reporting, improving continuity of care and operational tracking.

Necessity of the Database

A relational database is crucial for this project because the system's core functionalities depend on storing long-lived clinical data and enforcing rule-based workflows across multiple users and branches. Appointment scheduling, medical recordkeeping, prescriptions inventory tracking, billing generation, and vaccination tracking all require consistent relationships, integrity guarantees, and efficient querying. Therefore, the database is not only used for persistence, but also as the primary mechanism to validate constraints, maintain the pet's medical timeline, and support cross-branch operations while ensuring the consistency of medical care.

a. Persistent Clinical Data

- Pet profiles (species, breed, age, allergies) must be stored permanently.
- Each visit must be recorded so that veterinarians can later review:
 - past diagnoses
 - prescriptions
 - vaccines
 - vaccination records and vaccination plan
 - allergy information
- This enables a consistent medical timeline for every pet across repeated visits.

b. Appointment Booking Requires Reliable, Shared State

When a pet owner selects a vet and a time slot, the system must query the database to:

- detect scheduling conflicts,

- enforce maximum daily appointment limits for the veterinarian,
 - check whether the pet has outstanding unpaid bills,
 - then confirm the appointment or return an appropriate rejection message.
- These validations require persistent appointment and billing records and must stay correct under concurrent use.

c. Data Integrity via Constraints/Triggers

The scenario expects the system to enforce critical rules at the data level:

- slot limits and conflict checks,
 - medication availability checks and stock deduction,
 - preventing negative stock, and enabling restock requests
 - generating billing records after visit completion,
 - maintaining timeline updates.
- These are best enforced with database mechanisms to guarantee correctness beyond the UI layer.

d. Querying and Reporting

A database is required to support frequent queries like:

- pet's medical history timeline,
 - vet availability queries for booking filters,
 - unpaid bill checks before booking,
 - inventory availability checks during prescription,
 - later-stage reports (e.g., inventory consumption, overdue vaccinations, cost breakdowns).
- These require structured data and efficient SQL queries.

e. Additional Functionality Support (Microchip Registry)

If the “Pet Microchip & Lost/Found Registry” feature is kept, a database is needed to:

- store microchip IDs with a uniqueness rule (at most one active chip per pet),
- store lost/found reports linked to pets/branches/users,
- allow cross-branch lookup of a scanned chip to identify the associated pet/owner quickly.

Target Users and Roles

1. Pet Owner

- Registers and updates pet profiles (species, breed, age, allergies)
- Books appointments

- Views medical summaries
- Views billing records
- Views vaccination plan
- Rates veterinarians
- Registers microchip information (additional functionality)
- Creates lost/found reports (additional functionality)

2. Veterinarian

- Views daily appointment schedule
- Accesses complete pet medical history
- Logs diagnoses, symptoms, treatments
- Prescribes medication (with inventory validation)
- Creates referrals to specialists at other branches
- Defines vaccination plans
- Finalizes appointments and generates visit summaries

3. Clinic Manager / Administrator

- Manages branch-level inventory
- Logs shipments and batch numbers
- Defines minimum stock thresholds
- Tracks expired/damaged items
- Monitors billing records
- Generates analytical reports:
 - Stock consumption
 - Vaccination compliance rates
 - Overdue vaccination statistics
 - Wasted inventory items due to expiration
 - Revenue per branch

1.2 System Requirements

Functional Requirements

The system shall provide at least the following functionalities:

1. **User Registration & Login:** Users can register and log in with a role (Pet Owner, Veterinarian, Manager/Admin).
2. **Role-Based Access:** Pages and actions are restricted based on user roles such as vets can create medical records; owners cannot.
3. **Pet Profile Management:** Pet owners can create/update pet profiles including species, breed, age, and allergies.

4. **Browse Branches & Veterinarians:** Owners can browse branches and veterinarians with filters (branch location, specialization, species expertise, available dates).
5. **Book Appointment with Validation:** Owners select a vet and a time slot; system checks scheduling conflicts, vet's maximum daily appointment limit, and unpaid bills; then confirms or rejects with a clear message.
6. **Open Appointment & View Medical History:** Veterinarian opens an appointment and views pet's complete medical history such as diagnoses, prescriptions, vaccinations, allergies.
7. **Create Medical Visit Record:** Vet logs diagnosis, symptoms, prescribed treatments, and follow-up notes for the appointment.
8. **Prescription & Inventory Deduction:** If medication is prescribed, system checks branch inventory availability and deducts stock accordingly.
9. **Referral Creation:** Vet can create a referral record linked to the pet and target branch/specialist.
10. **Billing Record Generation:** On completion of an appointment, the system generates a billing record (consultation + treatment + medication costs).
11. **Owner Visit Summary & Vet Rating:** Pet owner can view visit summaries and rate the veterinarian after the appointment.
12. **Inventory Management (Manager):** Manager views stock per branch with filters (medicine name, category, supplier, expiration status).
13. **Shipment Logging:** Manager logs new shipments with batch number, quantity, unit cost, supplier, and expiration date.
14. **Stock Thresholds & Waste Tracking:** Manager defines minimum thresholds; expired/damaged items are removed from available stock and waste logs are stored.
15. **Vaccination Plan & Records:** Vet defines vaccination plan; each vaccination record stores vaccine name, date, batch, next due date, and administering vet; owners see upcoming/overdue vaccines highlighted.
16. **Microchip Registration & Ownership Link:** Pet owners can register a microchip identifier for a specific pet (chip ID, implantation date, issuer/status).
The system enforces uniqueness so that a pet can have at most one active microchip in the registry and each chip maps to exactly one pet.
17. **Lost/Found Reporting Across Branches:** Owners can create a Lost Report for their pet (lost date, last known location, details, status).
Managers (or authorized staff) can create a Found Report when a microchip is scanned at a branch (found date/location, scanned chip ID, reporting branch), enabling cross-branch reporting.
18. **Cross-Branch Lookup & Reunification Support:** When a found microchip is recorded, the system can query the registry to identify the associated pet/owner and help initiate reunification steps (e.g., notifying the owner through stored contact information or showing the owner record to authorized staff).

Non-Functional Requirements

1. Usability

- All user types should be able to understand how to use the system without prior training.
- Core flows must require minimal and clear steps:
 - owners booking an appointment (browse/filter → select vet/slot → confirmation or clear rejection reason),
 - veterinarians documenting a visit (open appointment → view history → enter diagnosis/treatment → prescription/referral if needed → complete),
 - managers checking inventory and monitoring alerts.
- Search and filtering interfaces should be intuitive and provide immediate feedback.

2. Security

- The system must enforce role-based access control so that users can only access and modify data permitted by their role:
 - owners can manage only their own pets/appointments,
 - veterinarians can access clinical records for appointments they handle,
 - managers/admins can access operational dashboards.
- Sensitive data such as passwords and personal contact information must be stored securely and not exposed through unauthorized queries or pages.
- Clinical data and billing information must be protected against unauthorized viewing or modification.

3. Performance

- The system should respond quickly to frequent operations such as:
 - searching/filtering branches and veterinarians by location/specialization/species expertise/date,
 - loading a pet's medical history timeline (visits, prescriptions, vaccinations, allergies),
 - validating an appointment request (conflict checks, daily limit checks, unpaid bill checks).
- Typical workloads should be handled efficiently by proper indexing and query design, especially for history retrieval and filtered searches.

4. Reliability

- The database must remain consistent under concurrent usage (multiple owners booking, veterinarians updating records, managers logging shipments).
- Critical multi-step workflows must be transaction safe such as:

- prescription creation → inventory availability check → stock deduction → visit completion → billing record generation.
- The system should be able to recover from partial failures without leaving inconsistent states (a prescription saved but stock not deducted).

5. Data Integrity

- Referential integrity must be preserved using primary/foreign keys (appointments must reference existing pets/vets/branches; prescriptions must reference valid visits and inventory items).
- Constraints and triggers should enforce key business rules:
 - no overlapping/conflicting appointments for the same vet and time slot,
 - maximum daily appointment limit for a vet,
 - no negative inventory after prescription deductions,
 - billing record generation on visit completion.
- The system should prevent invalid or incomplete records (orphan visits, prescriptions without associated visits).

1.3 System Limitations

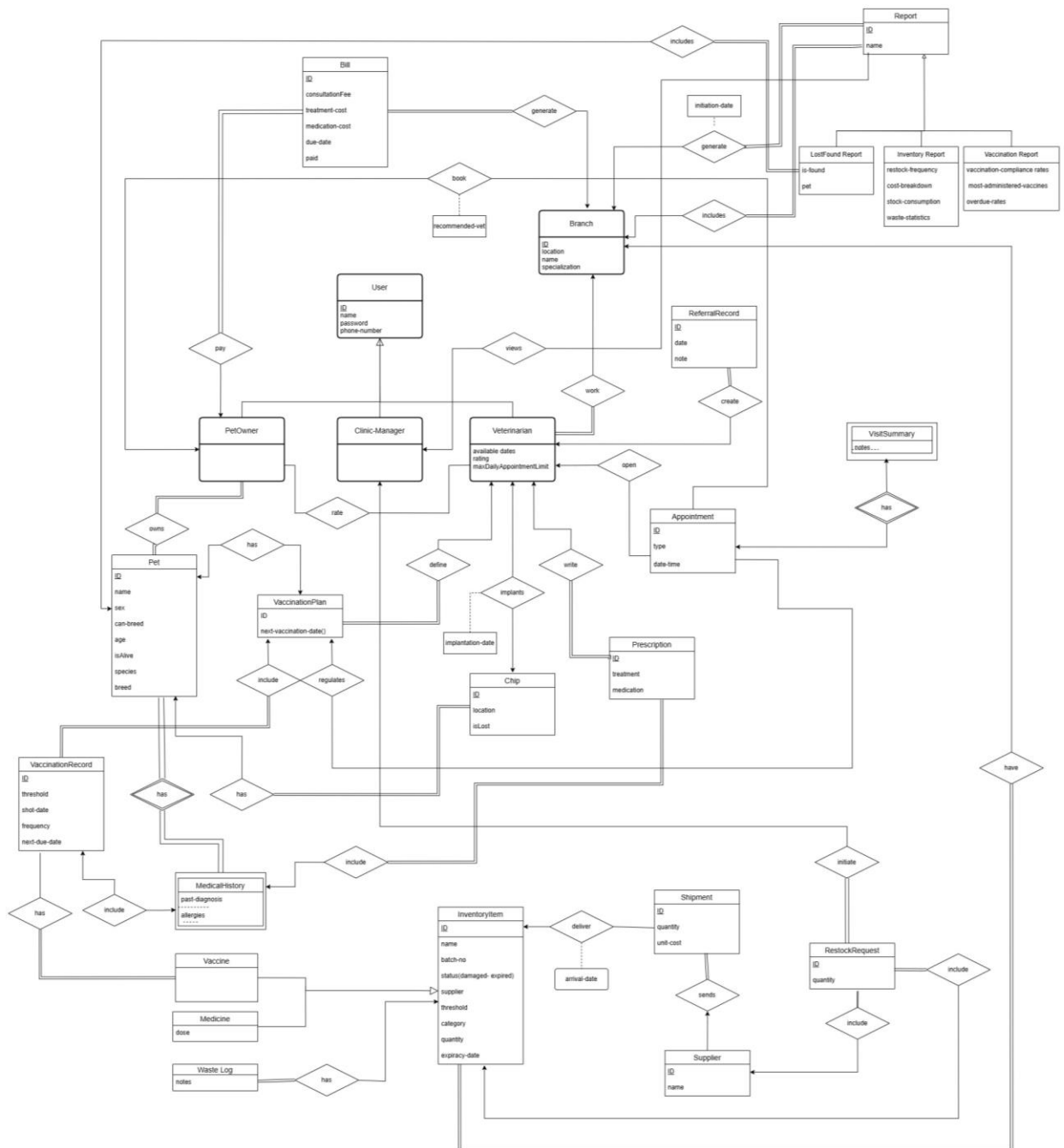
- Each user must have a unique user identifier and a unique email.
- Each account must belong to exactly one role (Pet Owner, Veterinarian, or Manager/Admin).
- Each pet must be registered by exactly one pet owner (a pet cannot belong to multiple owners).
- A pet owner can have multiple pets.
- Each appointment must be associated with exactly one pet, one veterinarian, and one branch.
- An appointment must have a scheduled date/time and a status such as scheduled, completed, and canceled.
- A veterinarian cannot have overlapping appointments.
- Each veterinarian has a maximum daily appointment limit, and the system must reject bookings that exceed this limit.
- A booking must be blocked if the corresponding pet (or its owner) has outstanding unpaid bills.
- Only the assigned veterinarian can finalize/complete an appointment.
- A medical visit record can be created only for an existing appointment.
- Each completed appointment should have at most one medical visit record (no duplicate visit records for the same appointment).
- Each prescription must be associated with exactly one medical visit.
- Each prescribed medication line must reference exactly one inventory item.

- Prescribing medication must trigger an inventory availability check at the appointment's branch.
- Inventory quantity must never become negative after prescription deductions.
- Each inventory item's stock is tracked per branch.
- Each shipment record must belong to exactly one branch and one inventory item and must store batch/expiration details.
- Expired or damaged items cannot be considered available stock and must be logged as waste.
- Each completed visit must generate exactly one billing record (consultation fee + treatment costs + medication costs).
- Each billing record must be linked to one medical visit and must have payment status (paid-unpaid).
- A pet owner can view only their own pets' billing records.
- Each veterinarian rating must be submitted by a pet owner for a completed visit.
- A pet owner can rate a veterinarian at most once per visit.
- Each vaccination record must belong to exactly one pet and store vaccine name, date administered, and next due date.
- Overdue/upcoming vaccination status is computed from stored dates and does not require manual marking.
- Each referral must be linked to exactly one pet and must include a target branch.
- A referral can only be created by the veterinarian handling the case.
- If the microchip module is kept, each pet may have at most one active microchip registered in the system.
- If the microchip module is kept, each lost report must reference exactly one pet and be created by the pet's owner.
- If the microchip module is kept, each found report must reference exactly one microchip and include the reporting branch.

Out-of-Scope / Not Implemented

- The system will not implement real bank/credit-card payment processing; billing will be tracked internally as paid/unpaid.
- The system will not integrate with external national registries or third-party veterinary systems.
- The system will not provide video consultation features.

1.4 E/R Diagram



For detailed information regarding E/R diagram can be found in the following [link](#) .